

Multi-Agent design for lightweight forensic triage

PROBLEME DES TOKENS:

General idea:

To keep the forensic analysis pipeline within the token limits of today's LLM agents, we must aggressively filter the raw Volatility 3 output while still preserving all investigative leads. The key is to perform a **lightweight triage** first, identifying the 5–10 most suspicious artefacts, and then pull only the rows related to those artefacts from the larger files. Below are four concrete proposals.

BONUS METHOD:

Proposal C: Baseline Deviation & Whitelisting

Best when you have access to a known-good reference (gold image, a clean snapshot of the same OS and patch level, or the NSRL/authorized software list).

1. **Extract unique file paths from process/module lists** (`cut` the path from `pslist.txt` , `dlllist.txt` , `modules.txt`).
2. **Compare against a whitelist** – everything that remains is “unusual”. Even a simple list of legitimate Windows paths (`C:\Windows\System32\*` , `C:\Program Files\Microsoft\*`) will cut 95% of noise.
3. **Feed only the unknown paths** (and their associated PIDs) into the agent. If you have hash data, use `hashdump` module output and compare against NSRL.
4. **Do the same for handles, drivers, and callbacks** – remove all known-good binaries (`*.sys` in `driverscan` that are signed by Microsoft).

This reduces large files even before keyword grepping and can be fully automated with a simple script. The remaining “rare” items often map directly to the attacker's tools.

RESOUDRE LE PROBLEME DES HALUCINATIONS ET DE LA LIMITATION EN TOKEN:

The practical multi-agent design

- 1- **collector -> REMPLACER ça DIRECTEMENT PAR UN SIMPLE SCRIPT** qui donne un json initial à l'agent 1.

Runs the volatility commands and prepares the outputs.

- 2- **Agent 1: triage analyst**

Reads only the high-level file and returns:

- suspicious processes
- suspicious services/tasks
- suspicious paths
- ...

3- Script pour GREP. Envoie output à l'agent 2

4- Agent 2: pivot analyst

Receives only the output from grep and analyzes them:

- PID-based grep in cmdline, privileges, handles, dlllist, envvars
- path-based grep in filescan, dumpfiles, registry_printkey
- Examines if the suspicion is justified with this full context (needs intelligence -> task is not deterministic).

5- Agent 3: report writer

Correlates all pivots into a short incident narrative:

- initial access guess
- execution chain
- persistence
- credential access
- staging
- likely files of interest

That split keeps the token load low and makes the workflow repeatable.